- **R:** R is a powerful computer language for statistical computations and enriched graphical capabilities. It offers the typical range of standard statistical plots, including bar plots, histograms, pie charts, scatter plots, box plots, etc. The *ggplot2()* package of R is a very powerful visualisation package that uses the Grammar of Graphics concept. The latter is a framework to construct visualisation in a standard structured manner. The two major functions of visualisation in *ggplot2()* are *qplot()* and *ggplot()*. *qplot()* is similar to a standard *plot()* function for quick plot generation, while *ggplot()* provides granular control of everything in all categories of the graph. The other important package is *plot3D*, which contains several 3D functions. These are listed in Table 2.

3D function	Purpose
box3D	Draws the boxes between pairs of points.
image3D	Adds an image in a 3D image plot.
spheresurf3D	Plots a coloured image on a sphere
trans3D	Projects 3D elements to 2D.
hist3D	Generates a 3D histogram.
persp3D	Extends the 3D perspective plots defined by the *persp()* function.
scatter3D	Generates the scatter 3D plot.
surf3D	Plots the surface in 3D with colours.
rect3D	Draws a rectangle in 3D.
ribbon3D	Draws a perspective plot as ribbon.
points3D	Draws the coloured point in 3D.
lines3D	Draws the coloured lines in 3D.
isosurf3D	Plots isosurfaces from a 3D data set.
voxel3D	Plots isosurfaces as points.

Table 2: 3D based functions of the *plot3D* package

- **Orange:** This is a great visualisation toolbox for novices as well as experts, and is very simple to use. No prior programming experience is required. Orange also comes with a visual programming environment and its workbench consists of tools for importing data, dragging and dropping widgets, and links to connect different widgets for completing the workflow. Orange uses common Python open source libraries for scientific computing, such as Numpy, Scipy, and Scikit-learn, while its graphical user interface operates within the cross-platform Qt framework.
- **Facets:** Facets is an open source visualisation tool that can help you understand and analyse AI/ML data sets. It comprises two visualisations for understanding and analysing these: Facets Overview and Facets Dive. Facets Overview automatically gives users a quick understanding of the distribution of values across the features of their data sets. Multiple data sets, such as a training set and a test set, can be compared on the same visualisation. Facets Dive provides an easy-to-customise, intuitive interface for exploring the relationship between the data points across the different features of a data set. With Facets Dive, you control the position, colour and visual representation of each data point based on its feature values. If the data points have images associated with them, these images can be used as visual representations.
- **TensorWatch**: This is a splendid tool to visualise data streams. The main functionality of the tool is to monitor the process of learning models in real-time in Jupyter Notebook. TensorWatch allows the user to customise a part of the model, as well as customise how the user wants to visualise and create dashboards. TensorWatch supports different types of visualisations: direct graph, pie chart, histogram, scatterplot, and 3D versions of many graphs.

To conclude, data visualisations play a vital role in AI/ML projects. The best data visualisation tools include both OSV (open source visualisation) tools as well as proprietary licensed solutions. This article describes benefits of open source visualisation tools based on parameters like features, complex data handling, API support, ease of operations, development and update facility, etc. All the tools described in this article provide great functionalities for visualisation from the artificial intelligence and machine learning perspectives. END

References

[1] *https://github.com/facebookresearch/hiplot*
[2] *https://ai.facebook.com/blog/hiplot-high-dimensional-interactive-plots-made-easy/*
[3] *https://towardsdatascience.com/data-science-made-easy-interactive-data-visualization-using-orange-de8d5f6b7f2b*
[4] *https://ai.googleblog.com/2017/07/facets-open-source-visualization-tool.html*
[5] *https://medium.com/@vlatamas/tensorwatch-a-library-for-monitoring-and-visualizing-models-3ead33b557f6*

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